

Interpreting Safety: A LLM and STPA Approach

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Abstract. Artificial Intelligence (AI) models are increasingly used in complex systems such as autonomous vehicles (AVs), where safety and explainability are critical. However, existing explainable AI (xAI) methods focus on model-level transparency while neglecting system-level safety explanations (Gap 1), and prior applications of large language models (LLMs) in AVs often view the AV as a whole, overlooking potential risks arising from interactions among its internal components (Gap 2). To address these gaps, we propose a framework that integrates LLMs with System Theoretic Process Analysis (STPA), a structured method to analyse hazards and assess safety, to improve AV safety assurance. Our framework leverages LLMs for scenario analysis while incorporating STPA to identify unsafe control actions (UCAs) and filter them with real-world video data. We evaluated our method against Lingo-2 (a vision-language-action model developed by Wayve) in a simulated environment, demonstrating superior STPA-based explanations. To evaluate the framework, we employed two ground truth references for accuracy verification and conducted robustness testing, which outperforming traditional LLM-based explainers, as also confirmed by expert evaluations.

Keywords: STPA · LLM · Explainer · Safety

1 Introduction

Artificial Intelligence (AI) models are now widely deployed in complex systems across various domains, such as autonomous vehicles [35], healthcare [1,4], and finance [26]. These systems, known as learning-enabled systems (LESs), typically consist of hardware, software, and machine learning components. Ensuring safety is a critical requirement for LESs [2,3,12], as their decision-making processes can directly affect human life and property safety.

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Most AI models were used in LESs as black-box model [16,25,30], making explainability a crucial area of research. Safety in LESs encompasses multiple attributes, including explainability, robustness, security and privacy, etc. [17,31]. Safety in LESs involves several key attributes, such as explainability, robustness, security, and privacy, each contributing to overall trustworthiness [16]. Various explainable AI (xAI) approaches have been developed to enhance the transparency of AI models. Traditional approaches often rely on interpretable approximations of the original model. For example, Local Interpretable Model-Agnostic Explanations (LIME) uses local surrogate models to explain predictions [29,36], while SHapley Additive Explanations (SHAP) calculates feature importances via Shapley values to ensure reliable explanations [22]. Although current xAI techniques offer insight into how a model makes decisions, they often overlook system-level understanding, which considers how the AI model behaves and impacts the broader system once integrated into a LES, leading to *Gap 1: Existing xAI methods ignore the explainability of system-level safety.* In the domain of autonomous vehicles (AVs), large language models (LLMs) have been widely explored in xAI research. A key advantage of LLMs is their ability to understand scene information [6], allowing AVs to better account for safety while interacting with the environment and providing explanations to users or experts. For example, [23] developed a human-centred multimodal interpreter (HMI) system to incorporate human preferences, delivering visual, textual, and auditory feedback to enhance passenger trust in AVs. [9] integrates LLMs' language and reasoning capabilities into AVs to enhance and explain their decision-making process. The above examples focus on applying AVs in scenarios and developing new frameworks to enhance and explain their safety. They often treat the system as a black box, placing greater emphasis on interpreting the AV's decisions within its scenarios. However, they often overlook the safety of the internal components of AVs themselves, leading to *Gap 2: The potential risks between components within LES cannot be ignored.* Every interaction between AVs and their environment is inherently linked to the control and coordination of their internal components.

To bridge these gaps, we propose a framework that integrates LLMs with System Theoretic Process Analysis (STPA). This framework leverages the strengths of LLMs in natural language understanding and scenario analysis while incorporating STPA to improve safety assurance, which using STPA find the system-level risks. Through in-context learning, the LLMs are trained to internalise STPA principles and domain-specific knowledge, enabling them to identify unsafe control actions (UCAs) and analyse control structures with a high degree of precision. Additionally, the framework utilises video (sequential image) data as real-world evidence to validate and filter UCAs, ensuring that the identified hazards are grounded in observable scenarios. This process allows the framework to systematically identify potential loss (safe) scenarios by correlating safety analyses with empirical data.

We designed a user case to demonstrate our framework and simulate real-world scenarios in a simulator. To evaluate its effectiveness, we compared it with Lingo-2[1], a market-available model. The results showed that our framework not

[1] https://wayve.ai/thinking/lingo-2-driving-with-language/.

only meets traditional AV requirements for scenario safety but also provides STPA-based explanations for experts. Additionally, we conducted accuracy and robustness experience compared with the ground truths. The results indicated that our framework outperforms LLM-based interpreters in terms of traceable explanations and accuracy. By integrating LLMs with STPA, we directly address *Gap 1* by extending explainability to system-level safety constraints, and *Gap 2* by explicitly capturing and explaining risks that emerge from inter-component interactions within LES. Finally, we validate that our framework operated on multiple platforms and produces consistent results. The contributions of this paper are as follows: We propose a framework that integrates LLMs and STPA to explain AV explainability in scenarios for safety experts by accounting for timesteps to enable more intuitive identification of UCAs, with the framework accessible at https://github.com/YiQi0318/LLM-based-STPA-Interpreter.

2 Background and Related Work

2.1 Large Language Model as Explainer

LLMs have significantly advanced the field of autonomous driving by providing natural language explanations for complex decisions, thus improving transparency, trust and user interaction [11,27]. For example, why automotive vehicles stopped or selected a particular route, the explanation bridging the gap between opaque AI systems and user understanding [23]. [9] proposed a framework leverages LLMs to enhance contextual understanding by integrating diverse sources, such as traffic regulations and accident reports, ensuring that decision-making prioritises both safety and regulatory compliance. [24] introduces a benchmark for autonomous driving visual question answering, enabling users to interactively assess the environment in which the autonomous system operates. By leveraging the capabilities of LLMs [5,8], these systems can better articulate the reasoning behind their actions in various driving situations, providing greater transparency and trust. Unlike these methods, our framework emphasises explaining the system-level (un)safety of autonomous vehicles with STPA in scenarios, which STPA constructs control structures within AVs to identify UCAs, effectively bridge the Gap 2. It is specifically designed to assist safety experts in evaluating the potential risks that autonomous vehicles may encounter.

2.2 System Theoretic Process Analysis and Causal Analysis Based on System Theory

STPA, founded on the Systems-Theoretic Accident Model and Processes (STAMP) [19,21], is a systematic hazard analysis methodology aimed at identifying UCAs and causal factors within a system's control structure. By examining the system's control actions (CAs), STPA offers a structured framework to mitigate accidents. Its workflow involves four principal steps: defining system-level losses and hazards, modelling the control structures, identifying unsafe control

actions[2], and uncovering loss scenarios. STPA is widely used in the automotive industry [7] and in LES [28]. [13] using STPA to explain the relationships between different system components, enabling the identification of potential risks at the system level more effectively. As noted in [10,32], STPA helps identify key areas in a system that require improvement, thereby enhancing the ability of system to mitigate potential risks. Causal Analysis based on System Theory (CAST) like STPA, is used to analyse the causes of any loss. Its purpose is to prevent future losses by learning from past events. Unlike traditional methods that seek a single "root cause," CAST aims to uncover the full range of contributing factors by each component [20]. The CAST method is described in five stages, although these are not necessarily sequential [20,33]: 1. Assembling the basic information; 2. Modelling the safety control structure; 3. Analysing each component in the loss; 4. Identifying control structure flaws; 5. Creating an improvement programme.

3 Methodology

3.1 Framework

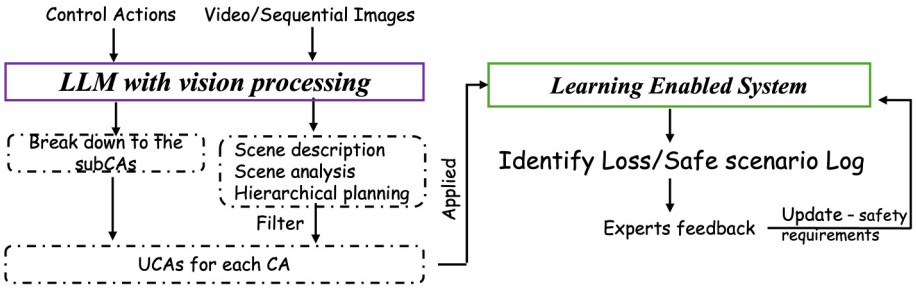


Fig. 1. Framework of LLM-based STPA Interpreter.

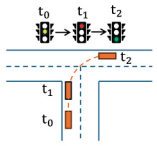
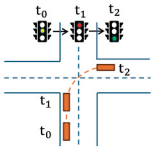
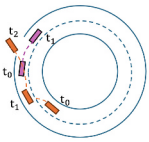
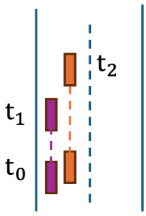
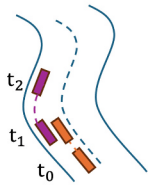
We employ an in-context learning approach to train the LLM with STPA knowledge, enabling it to function as an "interpreter" capable of identifying UCAs (cf. Fig. 1). The CAs and sensor inputs, such as video or sequential images from a camera, are fed into the interpreter. The interpreter break down the CAs to mapping the control structures, and generated UCAs. The interpreter also processes these images to provide detailed scene descriptions and analysis, referred to as "evidence." This evidence is used to filter UCAs, ensuring that only relevant and validated actions are considered. When applied to the LES, such as an AV, the interpreter facilitates the generation of a loss (or safe) scenario log based on real-world outcomes. This log is then used to update safety requirements and incorporate expert feedback, enhancing the overall safety and reliability of the system.

[2] UCAs have 4 actions' type, including provided, not provided, too early or too late, stopped too soon or applied too long.

3.2 Running Example

Baseline. We use a AV case showed in [18] as our baseline, adopting the control loop structure it defines. The AV in this context is an integrated system that combines hardware (e.g. Cameras, LiDAR, Radar, IMU & GPS), software (e.g. HCI), and machine learning components (e.g. Object Detection & Localisation) to enable autonomous driving capabilities.

Table 1. Potential Real-World Scenarios for Autonomous Vehicles (Orange Represents AV, Purple Represents Other Vehicle)

Attribution	T-junction	Crossroads	Roundabout	Highway	Country Road
Sketch Map					
Camera Record					
Simulation Scenarios					
Hazards Scenarios					

Scenarios. We developed sketch maps to represent various scenarios that an AV might encounter in real-world environments. As shown in Table 1, these scenarios include T-junctions, crossroads, roundabouts, highways, and country roads. Each column presents a different scenario, consisting of four components: a sketch map showing vehicle movements over time, real-world camera records (sourced from the WayveScenes101 dataset [37]), simulation scenarios recreated in the CARLA simulator, and hazard scenarios that include potential safety risks such as collisions. These scenarios serve to support the evaluation and improvement of AV safety under diverse driving conditions. This case study uses the AV driving under normal conditions as an example, focusing on the T-juction scenario described in Table 1. The scenario is described as "The AV is preparing to make a right turn at T-junction." The figure illustrates two distinct time steps, which can be broken down as follows: t_0 to t_1: The AV applies the brakes

as the traffic light changes from yellow to red. t_1 to t_2: The AV begins steering as the traffic light changes from red to green.

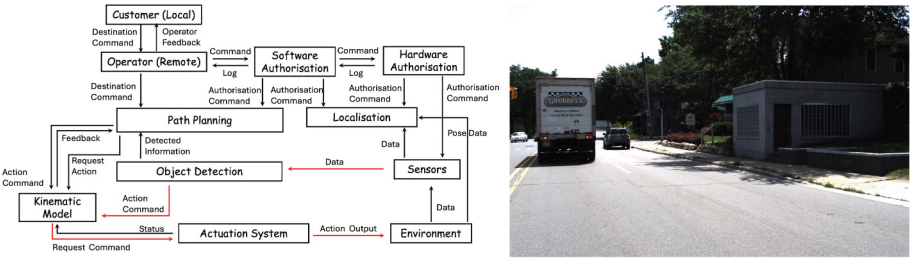


Fig. 2. Left: Control Structure Adapted from [18]. Right: Camera Recording from [14].(Color figure online)

Identify Control Actions. As shown in Fig. 2 (Left), the STPA control structure[3] outlines a sequence of coordinated CAs required for the AV to complete a right turn. Between t_0 and t_1 (Red line), the AV's onboard camera continuously captures real-time environmental images, which are fed into the object detection model for analysis. This model identifies relevant objects, such as pedestrians, other vehicles, and traffic signals, and determines potential hazards in the scene. Upon detecting a need for deceleration—such as recognising a red traffic light or an obstacle—the model generates a brake command. This brake command is transmitted to the kinematic model, which translates the decision into specific control signals. The kinematic model then forwards the braking request to the actuation system, which physically engages the braking mechanism, ensuring that the AV responds appropriately to its surroundings.

Identify Unsafe Control Actions. Following the standard STPA steps, we identified system-level hazards for the AV, including *[H1]: Loss of property* and *[H2]: Human damage*. Based on the CAs, we derived UCAs. During the identified CA from t_0 to t_1, where the camera captures environmental images for the object detection model, for example, potential UCAs shown in Table 2 include: 1) the camera providing blurry or poor-quality images, leading to incorrect object detection (H1, H2), and 2) the camera failing to capture images due to a malfunction, leaving the system without critical data (H1, H2). These UCAs highlight risks that could compromise the AV's safety and performance.

Vision Processing. When we give a video or sequential images, the LLM will be give related information includes three parts: 1) Scenarios Description (environment description, e.g. weather, time, road and critical object identification), 2)

[3] In STPA, the control structure refers to the relationship between the different components of a system—specifically, how controllers (such as an autonomous vehicle's control system) interact with processes, sensors, and actuators to influence the system's behaviour. The goal is to analyse how these components work together and to identify potential UCAs that could lead to system failures or hazards [21].

Table 2. UCAs for the CA: *Camera captures environmental images for the object detection model*

UCAs	Description
1.1	The camera provides the environment image, but it is blurry or of poor quality, causing the object detection model to fail or produce incorrect results (H1, H2).
1.2	The camera provides an incomplete environment image, omitting critical areas or objects in the scene (H1, H2).
1.3	The camera provides an image when the environment is not properly illuminated, leading to detection errors (H1, H2).
2.1	The camera does not provide data due to a malfunction or system error, leaving the object detection model without necessary data (H1, H2).
3.1	The camera captures the environment before the AV reaches the area of interest, leading to outdated or irrelevant input to the object detection model (H1, H2).
3.2	The camera captures the environment too late, after the AV has already passed critical points, causing delays in decision-making or execution (H1, H2).
4.1	The camera stops capturing the environment prematurely, missing critical parts of the scene during the AV's operation (H1, H2).

Scenarios Analysis (static attributions, e.g. building, traffic lights; motion states, e.g. particular behaviours), 3) Hierarchical Planning (e.g. actions decision).The scenario-related information was extracted from Fig. 2 (Right), which visually represents the environment of AV and associated factors. To demonstrate this process in our case study, we selected t_0 as a representative time step and input it into the LLM for further analysis. This allows us to examine how the LLM generates scenario details and refines the decision-making process in AV applications. In the t_0, for the scenarios description, scenarios analysis and hierarchical planning is shown in Table 3.

Unsafe Control Actions Filter. In the visual evidence of Timestep t_0, we observe the following: 1) the traffic light is currently yellow, 2) indicating the need to prepare for deceleration or complete stop, and 3) the truck in the centre lane is slowly slowing down, while the grey vehicle in the adjacent lane is maintaining a slow or moderate pace. To ensure relevance of UCAs in t_0, we apply the measure of Cosine similarity, in which the cosine similarity between two vectors A and B is computed as:

$$\text{sim}(A, B) = \frac{A \cdot B}{\|A\|\|B\|}$$

Table 3. Evidence of Scenarios Information

Category	Description
Weather	Clear, sunny
Time	Daytime with natural light
Road	Two-lane, clear markings, side-walk
Lane	Double yellow, white edge lines
Object	Truck, gray vehicle
Traffic Light	Amber signal
Building	Gray, glass blocks (right side)
Motion	Truck (Centre line): Decelerating Gray vehicle (Right line): Slow pace
Behaviour	Both vehicles preparing to stop for amber light
Actions	Decelerate, maintain safe distance
Decision	Slow down, monitor braking

where $A \cdot B$ represents the dot product of A and B, and $\|A\|$ and $\|B\|$ denote the Euclidean norms (magnitudes) of the vectors. The Cosine similarity quantifies the alignment between the two vectors, with a value closer to 1 indicating high similarity and a value closer to 0 indicating low similarity. which compares the alignment between scenario evidence and potential UCAs, filtering out low-similarity UCAs in t_0 and reducing non-relevant risks (cf. Algorithm 1).

Algorithm 1. UCAs Filter

Input: $evidence_info$, $UCAs$, $threshold$, LLM
Output: $filtered_UCAs$
Initialization $filtered_UCAs \leftarrow []$
$evidence_embedding \leftarrow$ `LLM.get_embedding(`$evidence_info$`)`
for $uca \in UCAs$ **do**
  $uca_embedding \leftarrow$ `LLM.get_embedding(`uca`)`
  $similarity \leftarrow$ `cosine_similarity(`$evidence$`,` $uca_embedding$`)`
  if $similarity \geq threshold$ **then**
    Append uca to $filtered_UCAs$
  end if
end for
return $filtered_UCAs$

Loss (or Safe) Scenarios Log. The AV experienced a crash at timestep t_2 due to risks that accumulated during earlier timesteps (t_0 and t_1). At t_0, initial risks such as blurry images and late capture (UCA1.1) delayed the AV's response to the yellow light, resulting in a delayed deceleration and an unsafe following distance behind the truck in the centre lane. By t_1, critical UCAs, including the failure to detect the red light and delayed braking, directly contributed to the

Table 4. Loss (or Safe) Scenarios Log

Timestep	**Relevant UCAs**	**Impact**
$\mathbf{t_0}$: Initial Risks Identified	**UCA1.1:** Blurry camera image, missing yellow light. **UCA3.2:** Late camera capture, delaying decision-making.	Delayed deceleration for yellow light. Unsafe following distance behind truck.
$\mathbf{t_1}$: Risks Worsened	**UCA1.2:** Camera missed red light. **UCA3.2:** Late brake command, delayed braking.	Failure to stop at red light. Risk to vehicles in adjacent lanes.
$\mathbf{t_2}$: Crash Occurs	–	Collision due to failure to stop at red light. Vehicle and property damage, potential injuries.
Causality :	• At (t_0), blurry image and late capture delayed response to yellow light, reducing stopping distance. • At (t_1), missed red light and delayed braking led to failure to stop at intersection. • These failures culminated in a crash at (t_2) at the intersection.	
Final Loss Scenario :	Unsafe actions at (t_0) and (t_1) led to a crash at (t_2) due to delayed braking.	
Timestep	**Mitigated UCAs**	**Outcome**
$\mathbf{t_0}$: Initial Safety Measures	**UCA1.1:** Clear camera image, accurate yellow light detection. **UCA3.2:** Timely camera capture, enabling prompt decision-making.	Accurate identification of yellow light. Safe following distance behind truck. Adequate reaction time for future hazards.
$\mathbf{t_1}$: Continued Safe Operation	**UCA1.2:** Camera captured red light. **UCA3.2:** Timely brake command.	AV decelerated smoothly to a stop. Safe distance from adjacent vehicles. Compliance with traffic rules.
$\mathbf{t_2}$: Safe Navigation Through the Intersection	–	Safe navigation through intersection. No collisions or unsafe conditions. Pedestrians remained stationary.
Causality :	• At (t_0), early detection of the yellow light and timely deceleration laid the foundation for safe operation. • At (t_1), recognition of the red light and appropriate braking ensured the AV came to a safe stop before the intersection. • At (t_2), the AV navigated through the intersection safely under the green light, avoiding hazards and maintaining compliance with safety requirements.	
Final Safety Scenario :	By properly mitigating UCAs and following effective decision-making at each timestep (t_0, t_1, t_2), the AV successfully avoided hazards, demonstrating safe and reliable operation throughout.	

AV's inability to stop at the intersection. These failures culminated in a crash at t_1, where the AV collided with another vehicle in the intersection (cf. Upper part of Table 4). In another potential result, the AV successfully avoided hazards and operated safely at t_2 due to effective risk mitigation during earlier timesteps (t_0 and t_1). At t_0, the AV accurately identified the yellow light and began decelerating promptly, maintaining a safe following distance behind the truck and setting a trajectory that ensured sufficient reaction time for future hazards. By

t_1, the AV recognised the red light and applied appropriate braking, coming to a safe stop before the intersection. Finally, at t_2, the AV navigated through the intersection safely under the green light, avoiding hazards and maintaining compliance with safety requirements (cf. Bellow part of Table 4).

4 Evaluation

4.1 Comparison

We use scene images from the WayveScenes101 Dataset [37] as an example. By inputting the same image into both Lingo-2 [24] and our framework, we compare their outputs as shown in Table 5. Unlike Lingo-2, our framework integrates STPA safety analysis, enabling the identification of Unsafe Control Actions (UCAs) for specific timesteps. We use UCA3.2 as an example, the AV captured the environment too late, leading to delays in decision-making or execution. Our framework still provides the fundamental information available in Lingo-2 (highlighted in teal). Additionally, because the "Scenery" section contains more detailed information, our model automatically infers relevant actions, covering Lingo-2's "Action" responses. For example, our framework also identifies the traffic light status and predicts the next action of the vehicle, preparing to slow down for the yellow light ahead. To illustrate safety information, the LLM-based STPA interpreter adds safety related explanation (highlighted in purple) to validate the system's behaviour under potential risks. Even though late perception was detected, the AV correctly recognised the yellow light, slowed down in time, and maintained a safe following distance in the centre lane, indicating no immediate hazard despite potential risks. We log these observations and share them with safety experts for validation, ensuring a continuous evaluation of safety performance at each timestep.

Table 5. Compared Results (First Row Represents the Lingo-2 Framework, the Second Row Represents the LLM-based STPA Framework.)

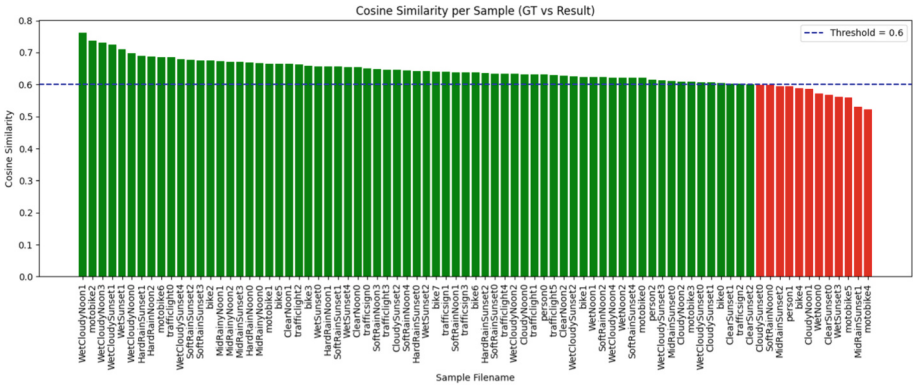
Video	STPA (UCAs)	Scenery	Action	Prediction
	*N/A*	*Can you list the traffic light you are observing? – I am observing the traffic lights are amber and red.*	*How are the amber traffic lights impacting your actions? – The amber traffic lights are causing me to decelerate and be prepared to stop if they turn red.*	*Can you turn right when in this lane? – No, I can go straight only.*
	*... **UCA3.2** The camera captures the environment too late, after the AV has already passed critical points, causing delays in decision-making or execution (H1, H2).*	*... Traffic Light (A traffic signal visible, showing a amber light, indicating preparation for stopping).*	*... Actions (Prepare to decelerate based on the amber light ahead. Maintain a safe following distance from the truck in the centre lane. Stay centred in the current lane. Maintain awareness of potential lateral movements from the vehicles in the adjacent lane.*	*1. The AV accurately identified the amber light and began to decelerate promptly. 2. A safe following distance was maintained be hind the truck in the centre lane.*

In the illustrated example, both frameworks correctly interpret traffic light status and infer actions such as deceleration or maintaining lane position. However, our framework enriches this interpretation by integrating context-specific safety reasoning. The "Scenery" and "Action" sections go beyond surface observations, inferring AV behaviour in response to road elements (e.g. traffic signals, nearby vehicles). The "Prediction" component further illustrates how the AV will likely respond over the next few moments, supporting safety analysis. Crucially, the LLM-based STPA interpreter adds expert-level safety commentary, validating that despite late perception, the AV maintained safe following distance and lane position, mitigating the potential hazard. The case study uses a relatively simple T-junction scene to illustrate our framework's core functionalities, it serves primarily as a proof of concept.

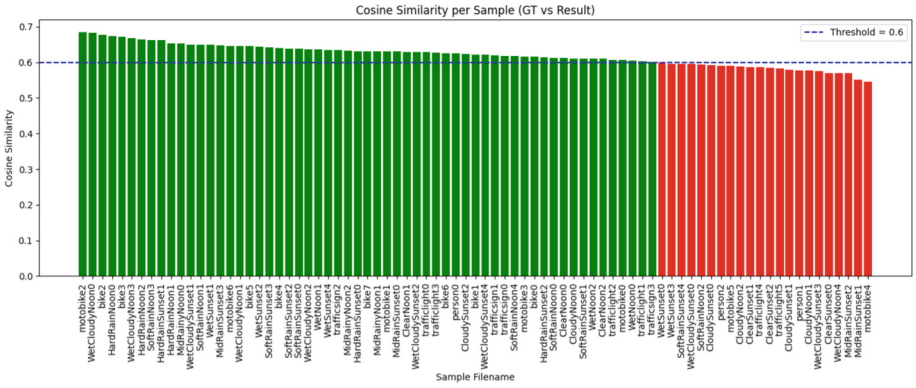
4.2 Accuracy

We modified the dataset from CARLA Steering DataSet [15] and Carla Computer Vision Project [34] as ground truths (GT) for comparison, which serves to provide both an internal consistency check and an external calibration benchmark. The first GT is generated by running our framework and then refining its outputs through expert standardisation. This version reflects how closely our automated system can approach the standards set by safety experts. The second GT is derived from the third step of the CAST method, which assess how well our framework aligns with external methodologies that may have different underlying principles or calibration procedures.

The first figure compares the output of our framework with the expert-standardised ground truth. Using a cosine similarity threshold of 0.6 (it balances sensitivity and specificity, where lower thresholds (e.g., 0.5) admitted semantically weak alignments and higher thresholds (e.g., 0.7 or 0.8) excluded many still acceptable explanations), most samples (in green of Fig. 3 (a)) show the alignment with expert expectations, confirming the internal consistency of our framework. Only a few samples fall below the threshold (in red of Fig. 3 (a)), suggesting isolated cases where fine-tuning may be needed, but overall performance remains solid. In contrast, the second chart assesses the same framework output but against the GT from CAST Step 3 (see Fig. 3 (b)). While many samples still meet or exceed the similarity threshold, there is a visibly higher number of red bars compared to the first chart. This suggests that the framework shows less agreement with CAST than with expert-standardised results. The divergence in performance likely reflects methodological differences between our framework and the CAST pipeline. Certain samples, especially those associated with more complex or dynamic conditions (e.g., MidRainSunset4, motorbike4), exhibit lower similarity scores, which likely reflects differences in methodology or calibration focus between our framework and CAST.



(a) LLM-based STPA Explainer v.s. Knowledge-based GT



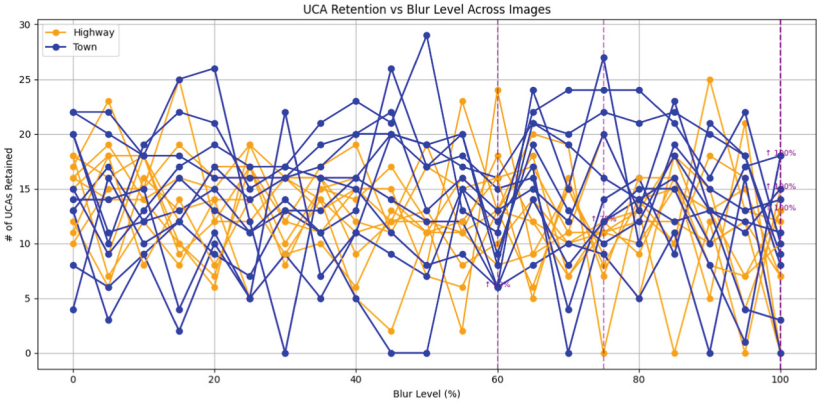
(b) LLM-based STPA Explainer v.s. Third Step of CAST

Fig. 3. Comparison between the LLM-based STPA Explainer and Ground Truths.(Color figure online)

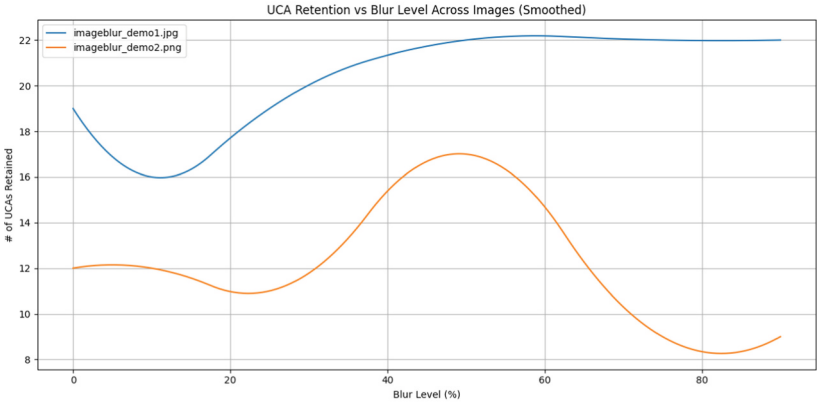
4.3 Robustness

We randomly selected 10 representative samples each from the Town and Highway scenes within our dataset (Sect. 4.2). These samples were then systematically subjected to varying levels of Gaussian blur, ranging from 0% (no blur) to 100% (maximum blur), to simulate real-world scenarios where camera motion or defocus can degrade image quality. The Fig. 4 provides a visualisation of how the system maintains performance under varying levels of image blur, a critical factor in real-world deployment where image quality can often be degraded. Robustness refers to the performance of the system to retain UCAs despite changes in input quality.

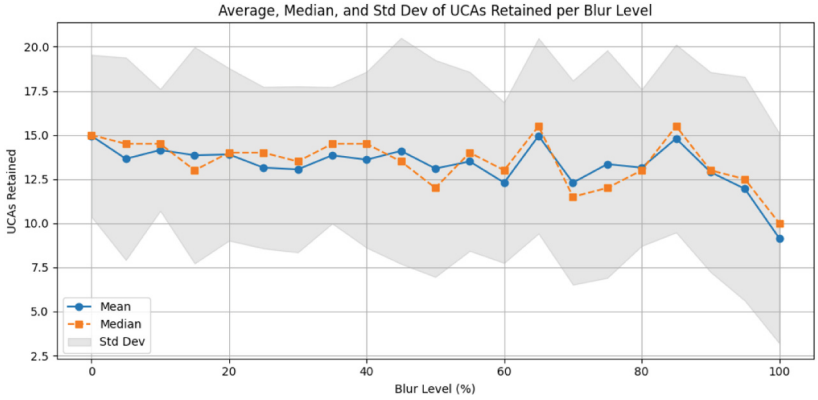
The Fig. 4 (a) presents the UCAs retention across a wide range of images from two environments (Highway and Town). Each line represents a single image,



(a) UCAs Retention v.s. Blur Level



(b) Single Sample UCAs Retention v.s. Blur Level (smoothed)



(c) UCAs Retention v.s. Blur Level (avg. & Med.)

Fig. 4. Comparison between the LLM-based STPA Explainer and Ground Truths.(Color figure online)

and the x-axis denotes increasing blur levels from 0% to 100%. Despite the visual variability, the retention patterns do not show a uniform decline as blur increases. Figure 4 (b) presents a smoothed view of UCA retention for two individual images. By applying curve fitting, this visualisation highlights the general trends in retention behaviour rather than moment-to-moment variability. After the blur level surpasses 20%, the mean number of retained UCAs begins to show an upward trend, suggesting a potential decrease in filtering stability under mild blur. For the Town scene (Blue), the mean UCAs retention stabilises once the blur exceeds 40%, although the overall filtering performance starts to decline beyond this point. In contrast, the Highway scene (Orange) exhibits considerable fluctuations in mean UCA retention after the 40% blur threshold, indicating reduced consistency and a more significant drop in the effectiveness of the UCA filtering under severe blur conditions.

Figure 4 (c) summarises the mean, median, and standard deviation of UCA retention across all blur levels. Both the mean and median remain relatively steady throughout most of the blur spectrum, indicating consistent performance across a wide range of conditions. However, as the blur level exceeds approximately 80%, a more noticeable decline in UCA retention emerges, signaling the point at which the UCA's filter performance begins to degrade. Despite this drop at extreme blur, the overall trend suggests that the system is resilient to moderate levels of blur, with performance degrading only under severe visual distortion.

5 Discussion

Our results show that integrating LLMs with STPA enhances system-level explainability in autonomous vehicles, outperforming traditional LLM-based methods in both structure and relevance of safety insights. This approach effectively bridges the gap between model transparency and operational safety by identifying unsafe control actions using real-world context. However, the framework has limitations. It currently depends on manually defined system models and hazard scenarios, which may restrict scalability to different AV architectures. The evaluation was also conducted in a simulated environment, which may not fully reflect real-world complexities.

6 Conclusion

We introduced a new framework that integrates LLMs with vision processing and STPA to enhance (un)safety explanation in real scenarios for autonomous vehicles. Our framework leverages the strengths of LLMs for scenario analysis, integrating them with STPA to systematically identify LES system-level risks and generate UCAs at various timesteps, using sequential video/image data as evidence, enhancing explainability in safety-critical environments. The proposed approach improves the ability of AVs facing the complex scenarios and offers transparent and explainable insights to safety experts.

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